



Real Estate Buyer Property Management System

Project Review – 3 | B.Tech Project | Database Systems Engineering and Distributed Backend Development

Academic Year: 2026–2027

V Vishwa Teja – 2520090033

B Sree Nikhil Reddy – 2520090058

Ch Vihamsh – 2520090090

Project Overview

What It Does

A web-based real estate platform that helps users discover properties, view property details, save favorite properties, and schedule property visits — all in one place.

Why It Is Needed

Property discovery, saving properties, and visit scheduling become inconvenient when information and communication are scattered across multiple channels.

Tech Stack

React.js • **Node.js + Express.js** • **MySQL** • **REST APIs**

Core Features



Property Browsing & Search



Wishlist Management



Visit Scheduling



User Dashboard



Property View Tracking



Contact Agent Functionality

Front-End Development

Built with **React.js** — a component-based UI framework enabling dynamic, responsive interfaces.



Component-Based UI

Reusable React components for consistent design across all pages.



Forms

Login, registration, property search, and visit scheduling forms.



API Integration

REST API calls to the Node.js backend for all data operations.



Navigation

React Router enables seamless page-to-page navigation without reloads.



Data Display

Property cards, details view, wishlist, and dashboard information panels.



Dynamic Rendering

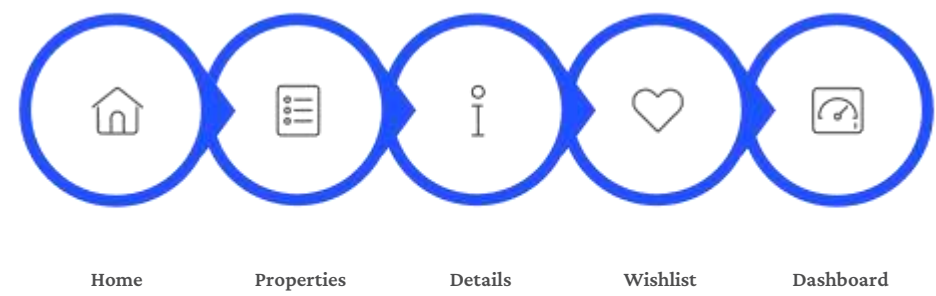
UI updates reactively using React state and effect hooks.

Frontend Modules & User Flow

Application Modules:

Home
Properties
Property Details
Wishlist
Dashboard
Login / Register
Property Search
Visit Scheduling
Contact Agent
About

Main User Flow:



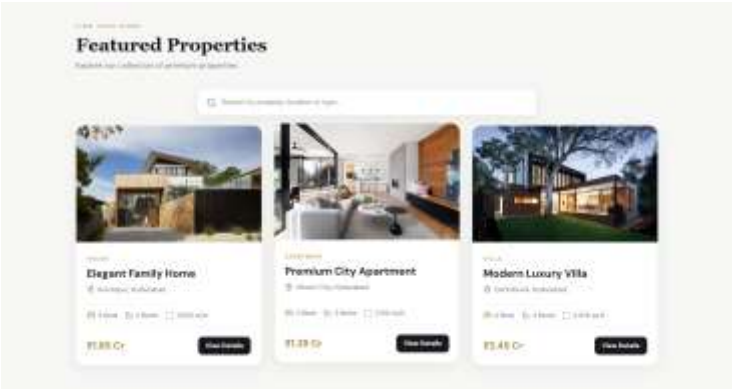
Frontend Screenshots

Key application pages — six primary views of the EstateLy platform interface.



Home Page

Property discovery and featured listings landing view.



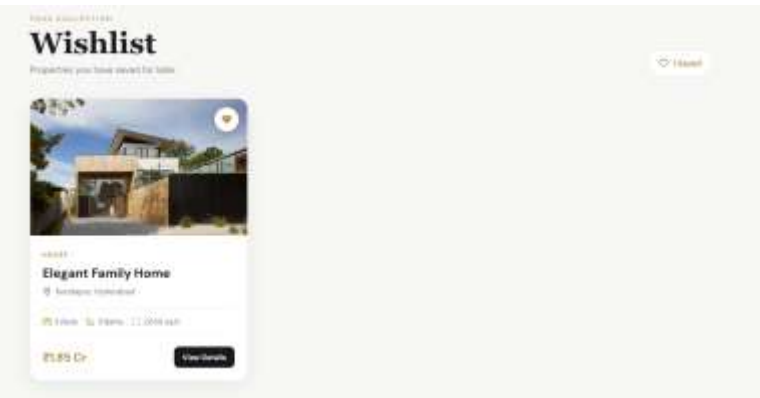
Properties Page

Property grid with search bar and filter controls.



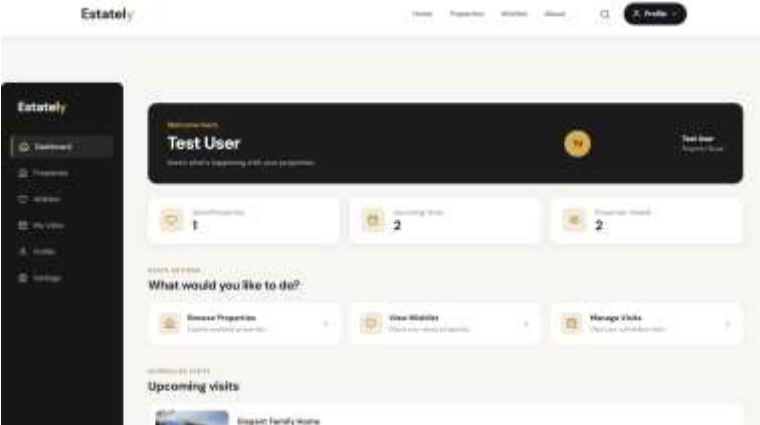
Property Details

Full property information, images, and agent contact.



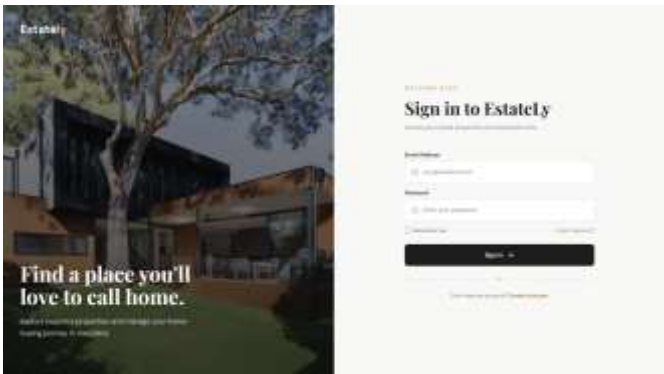
Wishlist

Saved favorite properties for quick access.



Dashboard

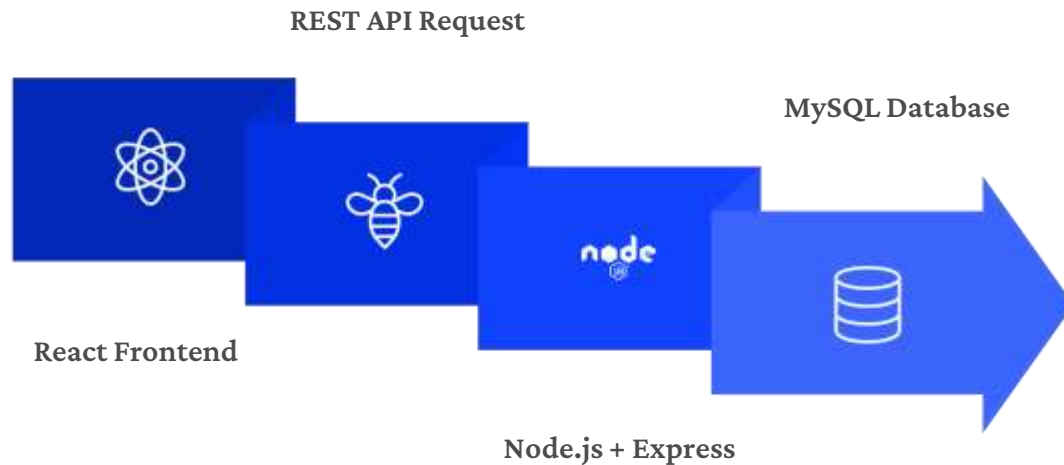
User profile overview and scheduled visit management.



Login / Register

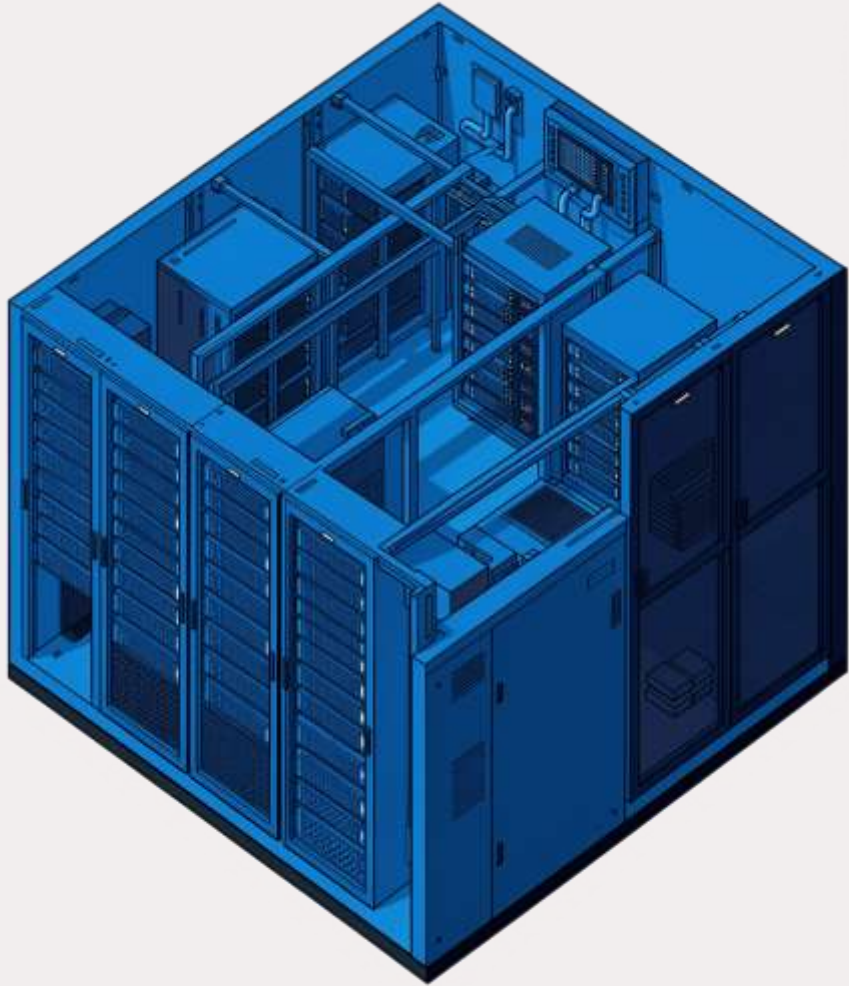
Secure user authentication and account creation pages.

Frontend–Backend–Database Flow



How It Works

- 1 User actions in React trigger REST API requests**
GET, POST, PUT, DELETE operations are dispatched based on user interaction.
- 2 Express routes handle requests and run SQL queries**
Node.js middleware validates input and queries the MySQL database.
- 3 Results return as JSON responses**
Structured data is sent back to the frontend over HTTP.
- 4 React updates state and re-renders**
UI reflects the latest data without a full page reload.



Database Design



Relational Approach

Data stored in structured, linked tables with well-defined relationships.



Purpose

Store users, properties, favorites, visits, and property-view information.



Data Consistency

Primary and foreign keys enforce referential integrity and valid data.

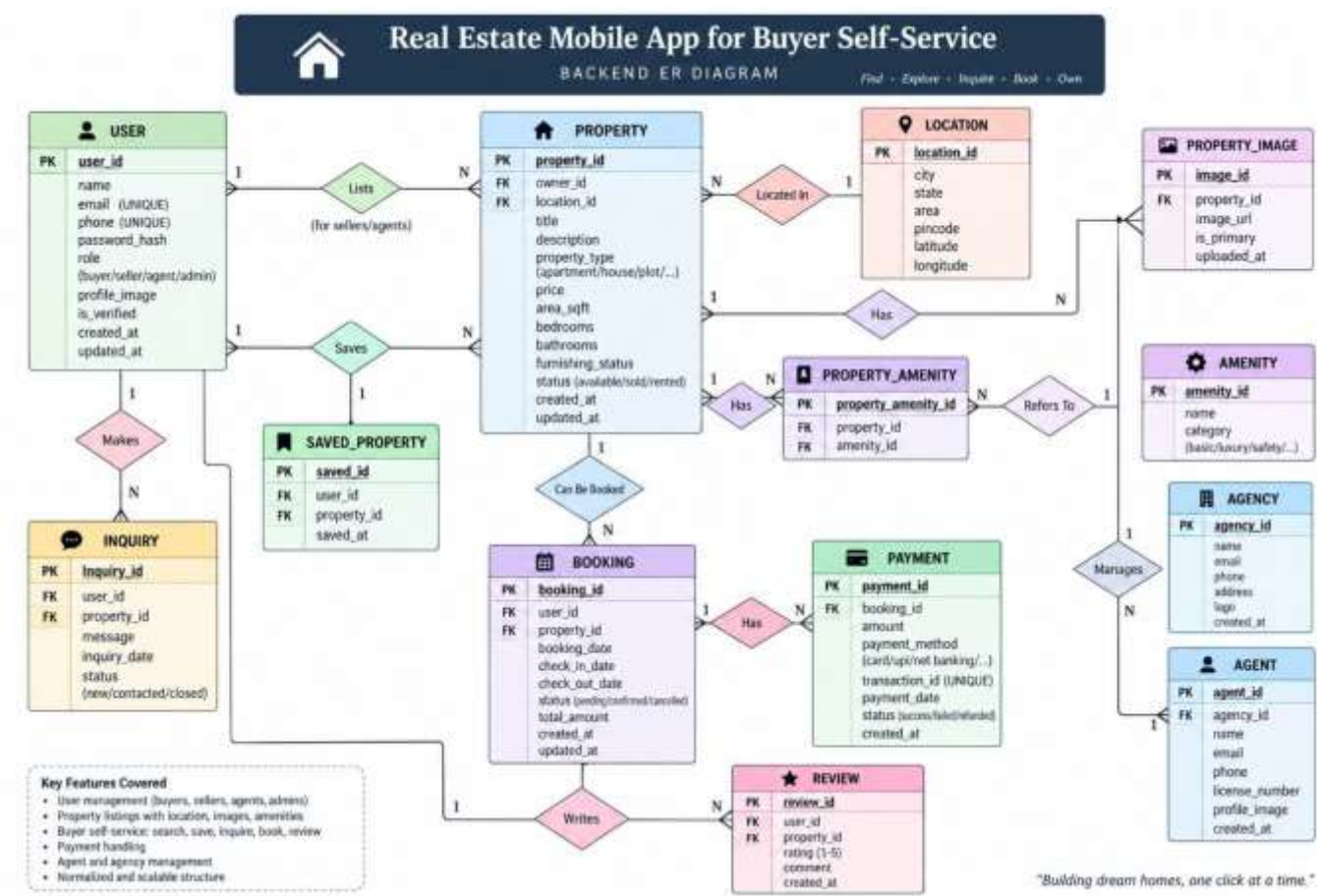


Reduced Redundancy

Each fact is stored once in its own table and referenced elsewhere.

ER Diagram — Database Schema

Five core tables with primary and foreign key relationships forming the EstateLy data model.



Tables and Their Purpose

Table	Purpose
users	Stores registered user information
properties	Stores available property listings
favorites	Connects users with their wishlist properties
visits	Stores scheduled property visits
property_views	Tracks properties viewed by users



Keys and Relationships

Primary Key

Uniquely identifies each record in a table.

Foreign Key

Connects records between related tables.

Referential Integrity

Keeps relationships between tables valid and consistent.

Why it matters

Maintains organized data and avoids invalid relationships.

Database Operations (CRUD)

CREATE

Add users, properties, favorites and visits.

READ

View properties, users, wishlist and visits.

UPDATE

Update user, property and visit information.

DELETE

Remove saved properties and records when required.

Sample SQL Queries

1. View all properties

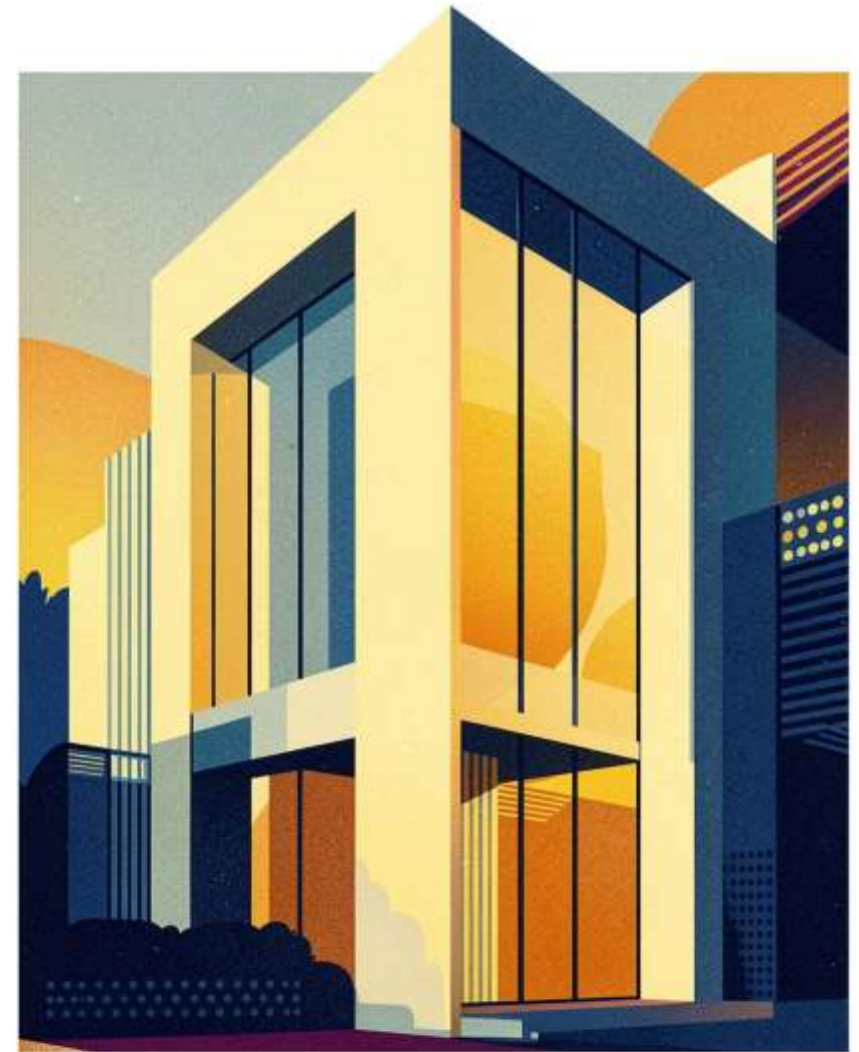
```
SELECT * FROM properties;
```

2. View all users

```
SELECT * FROM users;
```

3. View user's saved properties

```
SELECT properties.*  
FROM favorites  
JOIN properties  
ON favorites.property_id = properties.id  
WHERE favorites.user_id = ?;
```



Current Implementation / Progress

Frontend Development

- React frontend implemented
- Main UI pages developed
- Navigation and routing implemented
- Property search and filtering implemented
- Wishlist and dashboard implemented
- API integration completed

Backend Development

- Node.js + Express.js backend implemented
- REST API routes implemented
- User authentication implemented
- Wishlist and visit APIs implemented
- Property-view tracking implemented

Database

- MySQL database created
- 5 relational tables implemented
- Primary/foreign key relationships established
- CRUD operations tested

Conclusion and Next Steps

Conclusion

EstateLy combines a React frontend, Node.js + Express.js backend, and MySQL database to provide a centralized platform for property discovery, wishlist management, and visit scheduling.

Next Steps

- UI/UX improvements
- Advanced property search and filters
- Map-based property discovery
- Online property booking
- Automated visit reminders
- AI-based property recommendations



Thank You

